MyGradleApp

Step 1: Initialize the Gradle Application

Create a new directory named MyGradleApp and initialize a Java application using the gradle init --type java-application command. This sets up the basic boilerplate code and build wrappers.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine: ~/Devops/MyGradleApp
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops$ mkdir MyGradleApp
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops$ cd MyGradleApp/
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$ gradle init --type java-application
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)
Starting a Gradle Daemon (subsequent builds will be faster)

BUILD SUCCESSFUL in 2s
2 actionable tasks: 2 executed
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$
```

Step 2: Verify Initial Project Structure

Use the tree command to inspect the generated project layout. This confirms the creation of the src folder (containing App.java), the gradle wrapper, and the configuration files like build.gradle and settings.gradle.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$ tree
.
├── build.gradle
├── gradle
│   └── wrapper
│       ├── gradle-wrapper.jar
│       └── gradle-wrapper.properties
├── gradlew
├── gradlew.bat
├── settings.gradle
└── src
    ├── main
    │   └── java
    │       └── App.java
    └── test
        └── java
            └── AppTest.java

8 directories, 8 files
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$
```

Step 3: Execute Project Build

Run the gradle build command to compile the source code and run the default tests. This process downloads necessary dependencies (like Guava and JUnit) and creates the distribution files.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$ gradle build
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)
Download https://jcenter.bintray.com/com/google/guava/guava/23.0/guava-23.0.pom
Download https://jcenter.bintray.com/com/google/guava/guava-parent/23.0/guava-parent-23.0.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-annotations/1.14/animal-sniffer-annotations-1.14.pom
Download https://jcenter.bintray.com/com/google/code/findbugs/jsr305/1.3.9/jsr305-1.3.9.pom
Download https://jcenter.bintray.com/com/google/j2objc/j2objc-annotations/1.1/j2objc-annotations-1.1.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-parent/1.14/animal-sniffer-parent-1.14.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/mojo-parent/34/mojo-parent-34.pom
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_annotations/2.0.18/error_prone_annotations-2.0.18.pom
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_parent/2.0.18/error_prone_parent-2.0.18.pom
Download https://jcenter.bintray.com/org/codehaus/codehaus-parent/4/codehaus-parent-4.pom
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_annotations/2.0.18/error_prone_annotations-2.0.18.jar
Download https://jcenter.bintray.com/com/google/guava/guava/23.0/guava-23.0.jar
Download https://jcenter.bintray.com/com/google/code/findbugs/jsr305/1.3.9/jsr305-1.3.9.jar
Download https://jcenter.bintray.com/com/google/j2objc/j2objc-annotations/1.1/j2objc-annotations-1.1.jar
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-annotations/1.14/animal-sniffer-annotations-1.14.jar

> Task :test
WARNING: A terminally deprecated method in java.lang.System has been called
WARNING: System::setSecurityManager has been called by org.gradle.api.internal.tasks.testing.worker.TestWorker (file:/usr/share/gradle/l
ib/plugins/gradle-testing-base-4.4.1.jar)
WARNING: Please consider reporting this to the maintainers of org.gradle.api.internal.tasks.testing.worker.TestWorker
WARNING: System::setSecurityManager will be removed in a future release


BUILD SUCCESSFUL in 18s
7 actionable tasks: 7 executed
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$
```

Step 4: Inspect Build Artifacts

After a successful build, use the tree command again to view the build directory. This shows the generated .class files, the compiled .jar library, and the HTML test reports.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$ tree
.
├── build
│   ├── classes
│   │   └── java
│   │       ├── main
│   │       │   └── App.class
│   │       └── test
│   │           └── AppTest.class
│   ├── distributions
│   │   ├── MyGradleApp.tar
│   │   └── MyGradleApp.zip
│   ├── libs
│   │   └── MyGradleApp.jar
│   ├── reports
│   │   └── tests
│   │       └── test
│   │           ├── classes
│   │           │   └── AppTest.html
│   │           ├── css
│   │           │   ├── base-style.css
│   │           │   └── style.css
│   │           ├── index.html
│   │           ├── js
│   │           │   └── report.js
│   │           └── packages
│   │               └── default-package.html
│   ├── scripts
│   │   ├── MyGradleApp
│   │   └── MyGradleApp.bat
│   ├── test-results
│   │   └── test
│   │       ├── binary
│   │       │   ├── output.bin
│   │       │   ├── output.bin.idx
│   │       │   └── results.bin
```

Step 5: Run the Application

Execute the application using the gradle run command. This utilizes the application plugin to launch the main class, resulting in the "Hello world." output in the terminal.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$ gradle run
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)

> Task :run
Hello world.


BUILD SUCCESSFUL in 0s
2 actionable tasks: 1 executed, 1 up-to-date
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$
```

Step 6: Define a Custom Gradle Task

Open the build.gradle file using a text editor. Add a custom task named Hello using the doLast block to print a custom message when executed.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$ gedit build.gradle
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$ gradle Hello
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)

> Task :Hello
Hello!!


BUILD SUCCESSFUL in 0s
1 actionable task: 1 executed
```

```
 * This build file was generated by the Gradle 'init' task.
 *
 * This generated file contains a sample Java project to get you started.
 * For more details take a look at the Java Quickstart chapter in the Gradle
 * user guide available at https://docs.gradle.org/4.4.1/userguide/tutorial_java_projects.html
 */

// Apply the java plugin to add support for Java
apply plugin: 'java'

// Apply the application plugin to add support for building an application
apply plugin: 'application'

// In this section you declare where to find the dependencies of your project
repositories {
    // Use jcenter for resolving your dependencies.
    // You can declare any Maven/Ivy/file repository here.
    jcenter()
}

dependencies {
    // This dependency is found on compile classpath of this component and consumers.
    compile 'com.google.guava:guava:23.0'

    // Use JUnit test framework
    testCompile 'junit:junit:4.12'
}

// Define the main class for the application
mainClassName = 'App'

task Hello{
doLast{
        println("Hello!!");
}
}
```

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Step 7: Execute Custom Task

Run the newly created task using the command gradle Hello. The terminal will display the custom "Hello!!" message, demonstrating Gradle's extensibility.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$ gedit build.gradle
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyGradleApp$ gradle Hello
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)

> Task :Hello
Hello!!


BUILD SUCCESSFUL in 0s
1 actionable task: 1 executed
```